MUHAMMAD BAQER MOLLAH

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RESEARCH INTERESTS

• Connected Autonomous Vehicles • Critical Infrastructures • AI-Native Wireless • Multimodal Learning • Large Models • Edge Intelligence • Trustworthy AI • Privacy Aware Computing • Quantum Communications and Network • Quantum Safe Systems

CURRENT POSITION

University of Houston
Dept. of Information Science Technology
Lecturer August 2026 - Now
◦ Teaching Computer Information Systems (CIS) undergraduate courses in Fall 2026: *CIS 2334 Information Systems Applications* and *CIS 3343 Information Systems Analysis & Design*
◦ Developing lectures, course materials, assignments, and hands-on activities to facilitate student learning and apply information systems concepts to real-world problems
Post-Doctoral Fellow September 2025 - August 2026
Advisors: Prof. Kyu In Lee and Prof. Yunpeng (Jack) Zhang
Lab: Smart Systems Security Laboratory (SSS Lab)
Project: USDOT Tier 1 UTC Transportation Cybersecurity Center for Advanced Research and Education (CYBER-CARE)
◦ Worked closely with students to define research directions, plan project milestones, and oversee implementation and paper writing
◦ Designed and evaluated practical end-to-end security-aware mechanisms for connected vehicles
◦ Contributing to the writing of research proposals

EDUCATION

University of Massachusetts Dartmouth Fall 2022 - Summer 2025
PhD in Electrical and Computer Engineering
Advisor: Prof. Honggang Wang, IEEE Fellow (Currently at Yeshiva University, NY)
Dissertation: On Enabling Multi-Modal Sensing and Security Techniques for Connected Vehicles
Courses Taken (Selected): Network Security, Multimedia Communications, Advanced Computer Systems, and Dependable & Secure Computing

Jahangirnagar University, Dhaka, Bangladesh 2015 - 2016
MS in Computer Science
Research Project: Secure Data Sharing and Searching at the Edge of Cloud Connected Smart Devices
Courses Taken (Selected): Artificial Neural Networks, Embedded System Engineering, Computer Networks, Mobile & Wireless Communications, Information Theory & Coding System, Distributed Systems, and Cloud Computing

International Islamic University Chittagong, Chittagong, Bangladesh 2009 - 2014
BS in Electrical and Electronic Engineering
Final Year Project: E-Police System for Improved E-Government Services in Bangladesh
Courses Taken (Selected): Electrical Circuits and Electronics, Digital Electronics, Computer Programming, Communication Theory, Numerical Analysis, Signals and Systems, Digital Signal Processing, and Control System

ACADEMIC EXPERIENCES

University of Massachusetts Dartmouth September 2022 - August 2025
Dept. of Electrical and Computer Engineering
Teaching Assistant & Research Assistant
∘ Assisted in teaching *Introduction to Engineering & Computing (Lab)* with basic programming related projects with C++ in Arduino, guided them in using introductory engineering tools, and provided one-on-one mentoring whenever necessary to address diverse learning needs
∘ Helped manage lab sessions, monitored student progress, facilitated group discussions, and provided constructive feedback to support concepts
∘ Actively provided mentorship and guidance to the student teams to the semester-end group projects
∘ Conducted research in wireless sensing and communications for connected vehicles under the NSF funded project "Enabling Machine Learning based Cooperative Perception with mmWave Communication for Autonomous Vehicle Safety"

PUBLICATIONS

Google Scholar Profile: https://scholar.google.com/citations?user=VijQVZYAAAAJ&hl=en

* indicates mentored/co-mentored students

Submitted/Under Review

[4] **Muhammad Baqer Mollah**, Kyu In Lee, "Toward Secure Internet of Things through Access Control Encryption", *IEEE Consumer Electronics Magazine*, pp. 1 -13, 2026. (Submitted)

[3] **Muhammad Baqer Mollah**, Kyu In Lee, "Ensuring In-Vehicle Network Security with Edge Computing and Access Control Encryption", *IEEE Transactions on Consumer Electronics*, pp. 1 -13, 2026. (Submitted)

[2] **Muhammad Baqer Mollah**, Kyu In Lee, "Practical End-to-End Security-Aware Mechanism for Connected Vehicles by Multi-Level Access Control", *IEEE Transactions on Consumer Electronics*, pp.1-14, 2026. (Submitted)

[1] Muzamil Raheman Shaik*, **Muhammad Baqer Mollah**, Yunpeng Zhang, Lu Gao, Kyu In Lee, "EV Charging Network Security by Enhanced PKI with Decentralized Certificates Revocation", *2027 IEEE 45th International Conference on Consumer Electronics (IEEE ICCE)*, Las Vegas, NV, USA, pp. 1-6. (Submitted)

Journals/Magazines

[J5] **Muhammad Baqer Mollah**, Honggang Wang, Mohammad Ataul Karim, Hua Fang, "Multi-Modal Sensing and Fusion in mmWave Beamforming for Connected Vehicles: A Transformer Based Framework", *IEEE Transactions on Vehicular Technology*, vol. 75, No. 7, pp. 14937 - 14949, July 2026.

[J4] **Muhammad Baqer Mollah**, Honggang Wang, Mohammad Ataul Karim, Hua Fang, "Multi-Modality Sensing in mmWave Beamforming for Connected Vehicles Using Deep Learning", *IEEE Transactions on Cognitive Communications and Networking*, vol. 12, pp. 327-341, 2026.

[M2] **Muhammad Baqer Mollah**, Honggang Wang, Mohammad Ataul Karim, Hua Fang, "mmWave Enabled Connected Autonomous Vehicles: A Use Case with V2V Cooperative Perception", *IEEE Network*, vol. 38, no. 6, pp. 485-492, Nov. 2024.

[J3] **Muhammad Baqer Mollah**, Md Abul Kalam Azad, Yinghui Zhang, "Secure Targeted Message Dissemination in IoT Using Blockchain Enabled Edge Computing", *IEEE Transactions on Consumer Electronics*, vol. 70, no. 3, pp. 5389-5400, Aug. 2024.

[J2] **Muhammad Baqer Mollah**, Jun Zhao, Dusit Niyato, Yong Liang Guan, Chau Yuen, Sumei Sun, Kwok-Yan Lam, Leong Hai Koh, "Blockchain for the Internet of Vehicles towards Intelligent Transportation Systems: A Survey", *IEEE Internet of Things Journal*, vol. 8, no. 6, pp. 4157-4185, March 2021.

[J1] **Muhammad Baqer Mollah**, Jun Zhao, Dusit Niyato, Kwok-Yan Lam, Xin Zhang, Amer Ghias, Leong Hai Koh, Lei Yang, "Blockchain for Future Smart Grid: A Comprehensive Survey", *IEEE Internet of Things Journal*, vol. 8, no. 1, pp. 18-43, January 2021.

[M1] **Muhammad Baqer Mollah**, Md Abul Kalam Azad, Athanasios Vasilakos, "Secure Data Sharing and Searching at the Edge of Cloud-Assisted Internet of Things", *IEEE Cloud Computing*, Vol. 4, No. 1, January-February 2017, pp. 34-42.

Conferences

[C7] **Muhammad Baqer Mollah**, Kyu In Lee, "Multi-Level Secure Dissemination of Cooperative Awareness Messages in Internet of Vehicles," *2026 IEEE Future Networks World Forum (IEEE FNWF)*, Lisbon, Portugal, pp. 1-8.

[C6] Muzamil Raheman Shaik*, **Muhammad Baqer Mollahh**, Yunpeng Zhang, Lu Gao, Kyu In Lee, "Secure and Efficient Multicasting of Safety Messages for Connected Vehicles", *2026 IEEE Future Networks World Forum (IEEE FNWF)*, Lisbon, Portugal, pp. 1-6.

[C5] Mujtabaa Adam*, **Muhammad Baqer Mollah**, Kyu In Lee, "Mapping Extended Reality Threats in Connected Vehicles with Attack-Defense Tree", *2026 IEEE 15th International Conference on Consumer Electronics - Berlin (ICCE-Berlin)*, Berlin, Germany, 2026, pp. 1-6.

[C4] **Muhammad Baqer Mollahh**, George Zouridakis, Kyu In Lee, "Securing the Lifeline: Enforcing Read and Write Access Control in Vehicular Safety Services", *IEEE International Conference on Cyber Security and Resilience (IEEE CSR)*, Lisbon, Portugal, 2026, pp. 1-6.

[C3] **Muhammad Baqer Mollah**, Honggang Wang, Hua Fang, "Evaluating Vulnerabilities of Connected Vehicles Under Cyber Attacks by Attack-Defense Tree", *IEEE International Conference on Computing, Networking and Communication (ICNC)*, Maui, Hawaii, USA, 2026, pp. 1-6.

[C2] **Muhammad Baqer Mollah**, Honggang Wang, Hua Fang, "Multi-Modal Sensing Aided mmWave Beamforming for V2V Communications with Transformers", *IEEE Global Communications Conference (GLOBECOM)*, Taipei, Taiwan, 2025, pp. 1-6.

[C1] **Muhammad Baqer Mollah**, Honggang Wang, Hua Fang, "Position Aware 60 GHz mmWave Beamforming for V2V Communications Utilizing Deep Learning", *IEEE International Conference on Communications (ICC)*, Denver, CO, USA, 2024, pp. 4711-4716.

Book Chapter

[B1] **Muhammad Baqer Mollah**, Sherali Zeadally, Md Abul Kalam Azad, "*Emerging wireless technologies for Internet of Things applications: Opportunities and challenges*", In Encyclopedia of Wireless Networks, Springer, 2020, pp. 390-400, Editors: Prof. Xuemin Shen, Prof. Xiaodong Lin, and Prof. Kuan Zhang.

STUDENT MENTORING

Muzamil Raheman Shaik, Master's Student, University of Houston April 2026 – Present
Research Topic: Secure Multicasting for Connected Vehicles (Accepted to IEEE FNWF 2026)

Adam Mojtaba, Undergraduate Student, University of Houston January 2026 – Present
Research Topic: Threat Modeling and Security Evaluation for Extended Reality Systems (Accepted to IEEE ICCE-Berlin 2026)

PROPOSAL WRITING EXPERIENCE

◦ **Amazon Research Awards 2026:** Calibration-Free Transformer-Based Multi-Rate Latent Fusion for On-Device Tracking under Extended Occlusions, $80K, **PI:** Prof. Yunpeng Zhang, **Research Team Member:** Dr. Muhammad Baqer Mollah **Status:** Under review, **My Contribution:** Full proposal drafting and budget preparation.

◦ **University of Houston Internal Equipment Grant 2026:** Acquisition of Quantum-Aware AI and Quantum-Resilient Security Testbed for Future Connected Cyber-Physical Systems Research and Education, $50K, **PI:** Prof. Yunpeng Zhang **Status:** Under review, **My Contribution:** Full proposal drafting and budget preparation.

◦ **NSF - Collaborative Research:** CIF: Medium: Towards Fully Deep Learning-based mmWave Communications for Complex Mobile RF Environments, **PIs and Co-PIs:** Prof. Honngang Wang, Prof. Hua Fang, Prof. Hamid Sharif, UNL, **Status:** Not funded, **My Contribution:** One research thrust out of three in total.

RELATED TRAINING

CyberSTAR – CyberTraining for Secure Transportation and Reliable Autonomy June 22 to June 24, 2026
George Mason University, VA, USA
Supported by: National Science Foundation (NSF)

SKILLS

Linux/Unix based operating systems (e.g., Ubuntu, Kali), Embedded systems (e.g., RPi, Arduino), Networking tools, Virtualization, Ethical hacking, SQL, Python, MATLAB, Deep learning, PyTorce.

SERVICES

Web of Science Public Profile: https://www.webofscience.com/wos/author/record/T-4705-2019

Chair: Special Session on *Cyber Resilience for Trustworthy Connected Consumer Mobility* at IEEE 45th International Conference on Consumer Electronics (IEEE ICCE 2027), 8-11 January 2027, Las Vegas, NV, USA.

Reviewer for Journals/Magazines: IEEE Communications Magazine, IEEE Network, IEEE Wireless Communications Magazine, IEEE Wireless Communications Letters, IEEE Transactions on Mobile Computing, IEEE Transactions on Industrial Informatics, IEEE Transactions on Intelligent Transportation Systems, IEEE Internet of Things Journal, among others.

Technical Program Committee (TCP) Member: • IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (IEEE SmartGridComm) 2025 and 2026, • IEEE Vehicular Technology Conference (IEEE VTC) Fall 2026 and Spring 2027, among others.

EasyChair Webmaster: The 33rd International Conference on Computer Communications and Networks (ICCCN 2024), July 29 - 31, 2024, Big Island, Hawaii, USA.

MEMBERSHIP

Member of • IEEE • IEEE Communications Society • IEEE Vehicular Technology Society

WORK AUTHORIZATION

US Permanent Resident (green card holder), no employer sponsorship required.

REFERENCES

Available, upon request.